

USV Labs

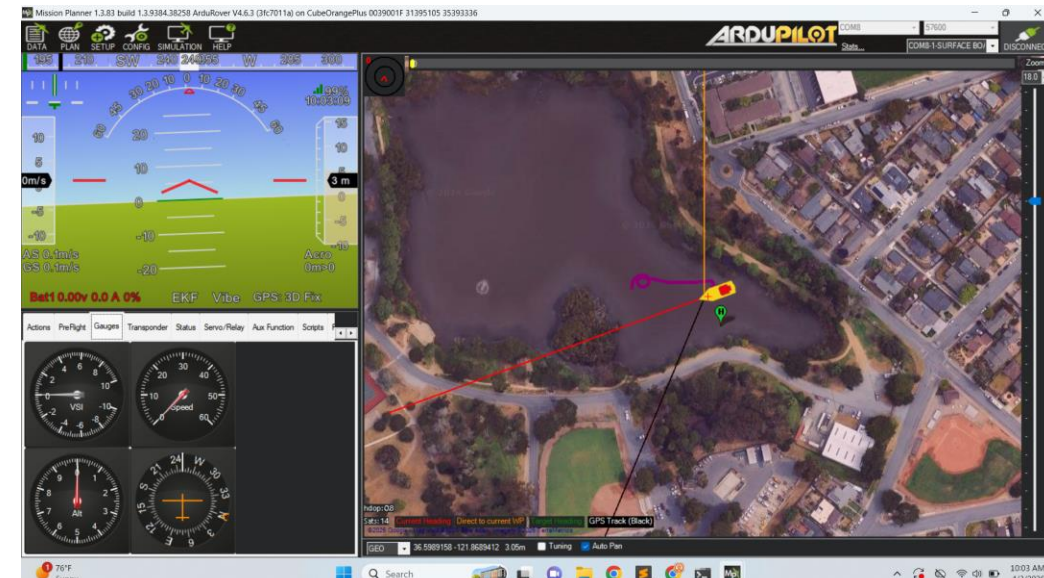
Objectives

Project:

Convert an R/C boat into a USV and navigate a course, consisting of a set of pre-defined waypoints, in minimum time.

Course:

Use the material from class to understand the design and implementation of the ArduPilot control system to tune the internal feedback control to maximize performance.



Week 1: Make it Work	• Build USVs. • Experiment: Manual control experiment to measure step responses. • Model the system response and find model parameters, based on experiment, to best represent the hardware.
Week 2: Make it Right	• Use ArduPilot documentation and model to simulate closed-loop control of the USV. • Use models to explore the design space. • Experiment: Tune ArduPilot PID gains.
Week 3: Make it Fast	• Competition week! (and write a report)

Lab 1: Measure open-loop response to identify USV model.

Mon/Tues in Lab:

- Build USVs, install Mission Planner, verify R/C control.

Wed/Thurs on Water:

- Log manual step response data.
- Plot step response in speed and yaw-rate.

Write-up due with next week's assignment.

